

Course: B. Tech. Branch: Chemical/Petrochemical Engineering Semester: VII

Subject Code & Name: BTCHC 703 (Transport Phenomena)

Marks: 60

Date: 01/02/2023

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected to answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

Q. 1 Solve any two of the following.		12
A) Summarize the procedure used in the solution of viscous flow problems by the shell balance method. What types of problems can be solved by this method?	CO1	6
B) How can we find out the average velocity, v_z for flow through a circular tube?	CO1	6
C) What is viscosity? Estimate the viscosity of N_2 at $50^\circ C$ and 854 atm, given $M = 28.0 \text{ g/g-mole}$, $p_c = 33.5 \text{ atm}$, and $T_c = 126.2 \text{ K}$.	CO2	6
Q.2 Solve any two of the following.		12
A) Show that the Hagen-Poiseuille formula is dimensionally consistent.	CO1	6
B) Heavy oil is passed through a pipe $5.08 \times 10^{-2} \text{ m}$ in diameter. the pressure drops over a pipe is 68958 N/m^2 . The viscosity of the oil is 200 cp & density is 800 kg/m^3 . The length of a pipe is 3.048 m. a) Calculate the volumetric flow rate of oil in lit/min b) Calculate and plot the momentum flux profile across the pipe.	CO2	6
C) Obtain an expression for the mass rate of flow w for an ideal gas in laminar flow in a long circular tube. The flow is presumed to be isothermal. Assume that the pressure change through the tube is not very large so that the viscosity can be regarded as constant throughout.	CO2	6
Q. 3 Solve Any two of the following.		12
A) Compare Fourier's Law of heat conduction with Newton's law of viscosity.	CO1, CO2	6
B) With a neat sketch derive the expression for the average temperature rise in heat conduction by an electrical heat source.	CO3	6
C) What is the friction Factor? What pressure gradient is required to cause diethylaniline, $C_6H_5N(C_2H_5)_2$, to flow in a horizontal, smooth, circular tube of inside diameter $D = 3 \text{ cm}$ at a mass rate of 1028 g/c at $20^\circ C$? At this temperature, the density of diethylaniline is $\rho = 0.935 \text{ g/cm}^3$ and its viscosity is $\mu = 1.95 \text{ cp}$.	CO2	6
Q.4 Solve Any two of the following.		12
A) Define heat transfer coefficients. Derive expression for it. Give the relation for flow around the submerged object	CO2	6
B) Distinguish between homogeneous and heterogeneous reactions. Which ones are described by boundary conditions in mass transport?	CO3	6
C) Oil is flowing in the laminar region in a $1.27 \times 10^{-2} \text{ m}$ dia. Tube at the rate of 22.72 lit/min. The oil viscosity is 300 CP and its density is 960.6 kg/m^3 . Calculate: a) Pressure drop per meter of pipe length. b) The wall stress, N/m^2 c) The velocity at the center of the tube.	CO2	6

d) Radial position at which velocity is equal to the average velocity. Given: $R = 0.635 \times 10^{-2} \text{ m}$

- Q. 5 Solve Any two of the following.** **12**
- | | | |
|--|-----|---|
| A) State and explain Fick's Law of Binary Diffusion. | CO2 | 6 |
| B) Define the Prandtl, Schmidt, and Lewis numbers. What ranges of Pr and Sc can one expect to encounter for gases and liquids? | CO3 | 6 |
| C) Calculate the mass flux of benzene through a layer of air 10 mm thickness at 25°C & 200 KN/m^2 (total pressure), partial pressure of benzene is 6 KN/m^2 , at the left side of the layer and 1 KN/m^2 at the right side. The mass diffusivity at this temperature & pressure is $4.4 \times 10^{-6} \text{ m}^2/\text{sec}$. | CO4 | 6 |

***** End *****